Qwen3

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> **Demonstration environment**

Development board: Jetson Orin series mainboard

Note: Due to the performance limits of the Jetson Orin NX, the small parameter version has to be run. The 8B qwen model cannot be used on a Jetson Orin NX with 8GB of RAM or less

Qwen 3 is the newest generation of large language models in the Qwen family, and it provides a complete set of dense and mixture-of-experts (MoE) models.

1. Model size

Model	Size
qwen3:0.6b	523MB
qwen3:1.7b	1.4GB
qwen3:4b	2.6GB
qwen3:8b	5.2GB

2. Performance

	Qwen3-235B-A22B MoE	Qwen3-32B Dense	OpenAI-o1 2024-12-17	Deepseek-R1	Grok 3 Beta Think	Gemini2.5-Pro	OpenAI-o3-mini Medium
ArenaHard	95.6	93.8	92.1	93.2	-	96.4	89.0
AIME'24	85.7	81.4	74.3	79.8	83.9	92.0	79.6
AIME'25	81.5	72.9	79.2	70.0	77.3	86.7	74.8
LiveCodeBench v5, 2024.10-2025.02	70.7	65.7	63.9	64.3	70.6	70.4	66.3
CodeForces Elo Rating	2056	1977	1891	2029	-	2001	2036
Aider Pass@2	61.8	50.2	61.7	56.9	53.3	72.9	53.8
LiveBench 2024-11-25	77.1	74.9	75.7	71.6	-	82.4	70.0
BFCL v3	70.8	70.3	67.8	56.9	-	62.9	64.6
MultiIF 8 Languages	71.9	73.0	48.8	67.7	-	77.8	48.4

1. AIME 24/25: We sample 64 times for each query and report the average of the accuracy. AIME'25 consists of Part I and Part II, with a total of 30 questions.
2. Aider: We didn't activate the think mode of Qwen3 to balance efficiency and effectiveness.
3. BFCL: The Qwen3 models are evaluated using the FC format, while the baseline models are assessed using the highest scores obtained from either the FC or prompt formats.

	Qwen3-30B-A3B MoE	QwQ-32B	Qwen3-4B Dense	Qwen2.5-72B-Instruct	Gemma3-27B-IT	DeepSeek-V3	GPT-4o 2024-11-20
ArenaHard	91.0	89.5	76.6	81.2	86.8	85.5	85.3
AIME'24	80.4	79.5	73.8	18.9	32.6	39.2	11.1
AIME'25	70.9	69.5	65.6	15.0	24.0	28.8	7.6
LiveCodeBench v5, 2024.10-2025.02	62.6	62.7	54.2	30.7	26.9	33.1	32.7
CodeForces Elo Rating	1974	1982	1671	859	1063	1134	864
GPQA	65.8	65.6	55.9	49.0	42.4	59.1	46.0
LiveBench 2024-11-25	74.3	72.0	63.6	51.4	49.2	60.5	52.2
BFCL v3	69.1	66.4	65.9	63.4	59.1	57.6	72.5
MultiIF 8 Languages	72.2	68.3	66.3	65.3	69.8	55.6	65.6

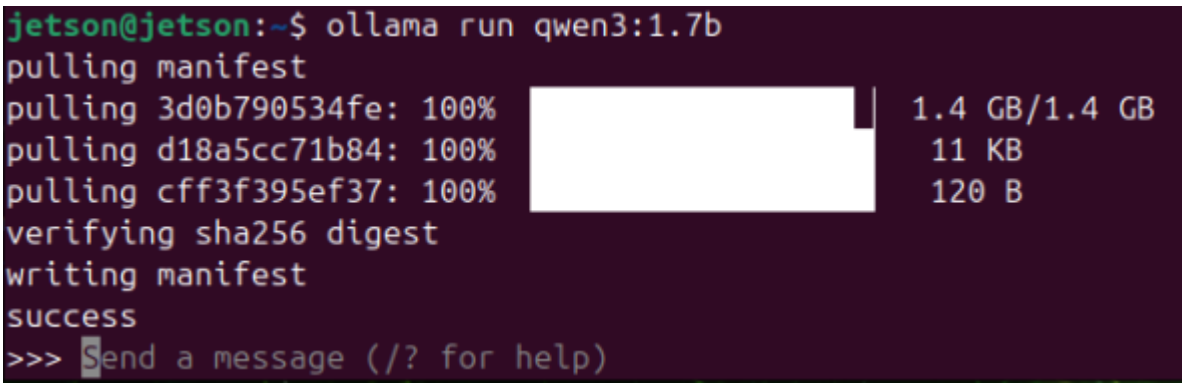
1. AIME 24/25: We sample 64 times for each query and report the average of the accuracy. AIME'25 consists of Part I and Part II, with a total of 30 questions.
2. Aider: We didn't activate the think mode of Qwen3 to balance efficiency and effectiveness.
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3. Using Qwen3

3.1 Run Qwen3

Use the run command to start the model; if the model has not been downloaded before, it is pulled automatically from the Ollama model library:

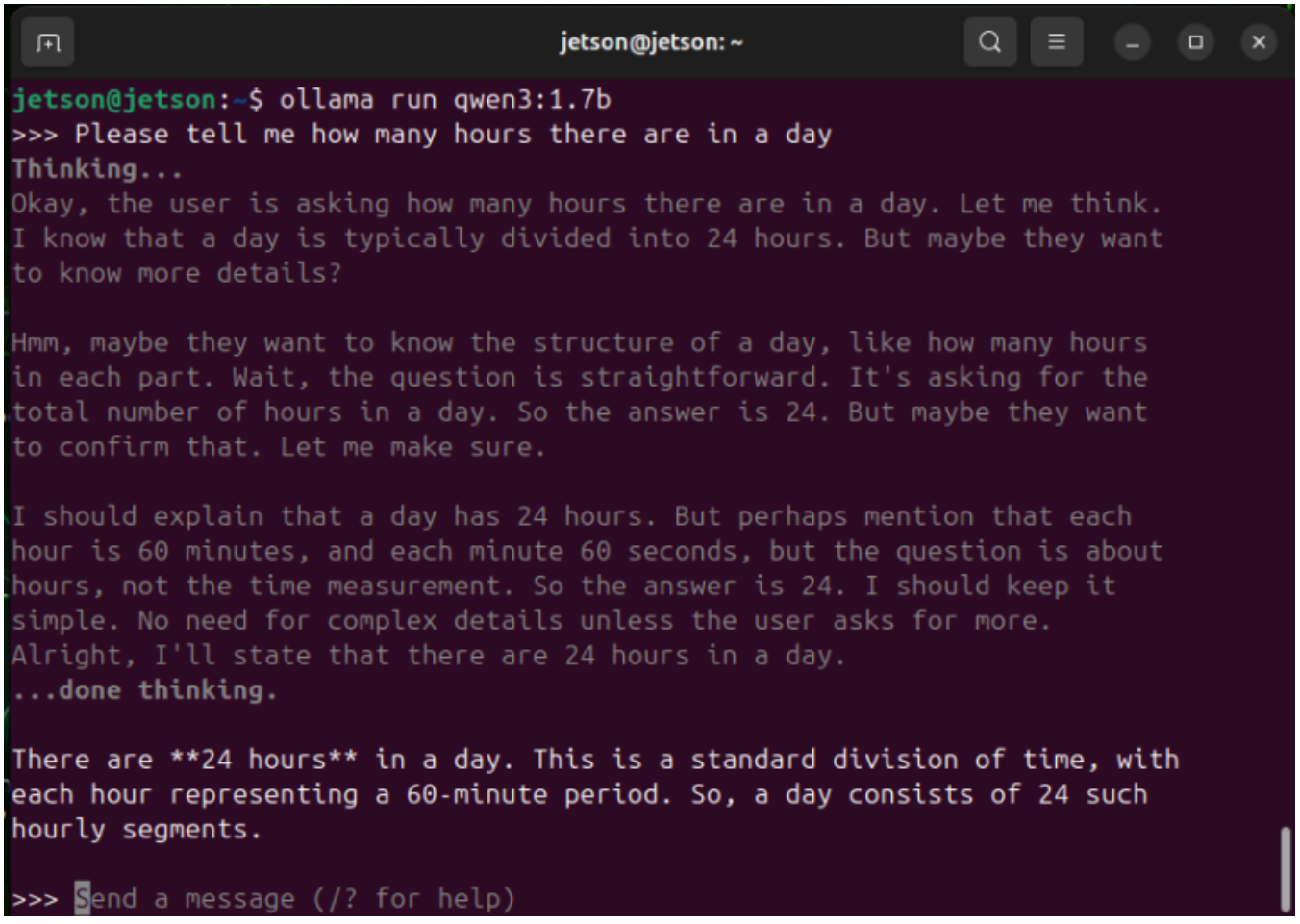
```
ollama run qwen3:1.7b
```



3.2 Start a conversation

```
Please tell me how many hours there are in a day
```

The time it takes to answer depends on the hardware configuration, so be patient!



3.3 End the conversation

Use the `Ctrl+d` shortcut or `/bye` to end the conversation!

References

> **Ollama**

Official website: https://ollama.com/

GitHub: https://github.com/ollama/ollama

> **Qwen3**

GitHub: https://github.com/QwenLM/Qwen3

Corresponding Ollama model: https://ollama.com/library/qwen3